Chapter 4. Classification: Logistic Regression, Softmax and SVM

Logistic regression, multiclass classification, support vector machines

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1 Odds, Logit and the Logistic Model

Motivation. For a binary label we need a model of a probability, a number in $[0,1]$, as a function of features. Modelling the log-odds as linear gives the logistic model, whose output is the sigmoid of a linear score.

For a probability p, the odds are $o = p/(1-p)$ ($p = 0.8$ gives odds 4) and

$$\operatorname{logit}(p) = \log \frac{p}{1-p} \in \mathbb{R}, \qquad \sigma(z) = \frac{1}{1+e^{-z}} = \operatorname{logit}^{-1}(z).$$

Definition 4.1: logistic regression

With augmented input $\mathbf{x} = (1, x_1, \ldots, x_d)$ and $y \in \{0,1\}$,

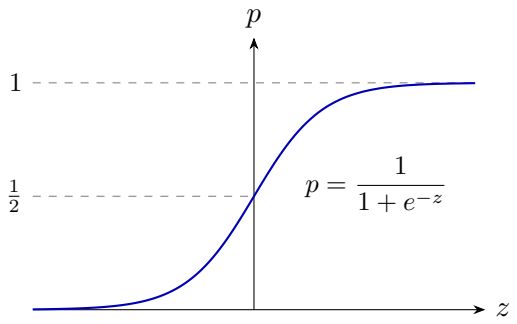
$$\log \frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}, \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

The score is linear and the probability transform nonlinear. Thresholding the probability at τ gives the hyperplane $\mathbf{w}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$ (at $\tau = 0.5$: $\mathbf{w}^\top \mathbf{x} = 0$).

Theorem 4.2: sigmoid identities

$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le \frac{1}{4}$, $1 - \sigma(z) = \sigma(-z)$, $\frac{d}{dz}\log(1+e^z) = \sigma(z)$.

Proof. $\sigma'(z) = e^{-z}/(1+e^{-z})^2 = \sigma(z)\cdot\frac{e^{-z}}{1+e^{-z}}$ and $\frac{e^{-z}}{1+e^{-z}} = \frac{1}{e^z+1} = \sigma(-z) = 1-\sigma(z)$. The maximum of $s(1-s)$ on $[0,1]$ is $1/4$ at $s = 1/2$. ■



The logistic function, inverse of the logit $z = \log \frac{p}{1-p}$: log-odds z are mapped to probabilities p.

2 Likelihood, Gradient, Hessian and Convexity

Motivation. Logistic regression has no closed form, so we must optimize numerically. The Hessian is positive semidefinite, so the loss is convex: every local minimum is global and gradient methods are reliable.

For data $(\mathbf{x}_i, y_i)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = p_i^{y_i}(1-p_i)^{1-y_i}$ and

$$J(\mathbf{w}) = -\sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] = \sum_i \left[\log(1+e^{z_i}) - y_i z_i\right], \quad z_i = \mathbf{w}^\top \mathbf{x}_i.$$

Theorem 4.3: logistic loss is convex

$\nabla J = \sum_i (p_i - y_i)\mathbf{x}_i = X^\top(\mathbf{p}-\mathbf{y})$ and $\nabla^2 J = \sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top = X^\top S X$, $S = \operatorname{diag}(p_i(1-p_i))$. Hence J is convex, and strictly convex if X has full column rank.

Proof. $\partial J/\partial z_i = \sigma(z_i) - y_i = p_i - y_i$ by the identities above, and $\partial z_i/\partial \mathbf{w} = \mathbf{x}_i$. Differentiating again, $\nabla p_i = p_i(1-p_i)\mathbf{x}_i$. For any $\mathbf{v}$, $\mathbf{v}^\top X^\top S X \mathbf{v} = \sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 \geq 0$; since $p_i(1-p_i) > 0$, the sum is positive whenever $X\mathbf{v} \neq \mathbf{0}$. ■

Example 4.4: one gradient step

$\mathbf{x} = (1, 2)$, $y = 1$, $\mathbf{w} = \mathbf{0}$: $p = 1/2$, $\nabla J = (p-1)\mathbf{x} = (-1/2, -1)$. With $\eta = 0.1$, $\mathbf{w} = (0.05, 0.1)$, score 0.25, $p \approx 0.562$: the probability of the correct label increased.

3 Labels in $\{\pm 1\}$, Logistic Loss and BCE

Motivation. The same model can be written with labels $\{0, 1\}$ (binary cross-entropy) or $\{\pm 1\}$ (logistic loss of the margin yf). The margin form connects logistic regression with the SVM and the other surrogate losses of this section.

With $Y \in \{\pm 1\}$ and score $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, $P(Y = +1 \mid \mathbf{x}) = \sigma(f)$ and $P(Y = -1 \mid \mathbf{x}) = \sigma(-f)$, so $-\log P(Y \mid \mathbf{x}) = \ell(Yf)$ with the **logistic loss** $\ell(z) = \log(1 + e^{-z})$. $Yf(\mathbf{x})$ is the *margin*: positive iff the sign of the score is correct.

Theorem 4.5: three names for one objective

Let $P(-1) = (1, 0)$, $P(+1) = (0, 1)$, $\mu(r) = (\sigma(-r), \sigma(r))$, $\tilde{Y} = (1 + Y)/2 \in \{0, 1\}$ and $\mathrm{BCE}(s, t) = -[t \log s + (1 - t)\log(1 - s)]$. For every $(\mathbf{a}, b)$ and every i,

$$D_{\mathrm{KL}}\big(P(Y_i) \,\|\, \mu(f(\mathbf{x}_i))\big) = \mathrm{BCE}\left(\sigma(f(\mathbf{x}_i)), \tilde{Y}_i\right) = \ell(Y_i f(\mathbf{x}_i)).$$

So KL minimization, BCE minimization and logistic regression are the same problem.

Proof. For a one-hot first argument, KL $= -\log$(model probability of the true class). If $Y = +1$ this is $-\log \sigma(r) = \log(1 + e^{-r}) = \ell(r)$, and BCE with $t = 1$ is $-\log s$. If $Y = -1$ it is $-\log \sigma(-r) = \ell(-r)$, and BCE with $t = 0$ is $-\log(1 - s) = -\log \sigma(-r)$. ■

Theorem 4.6: ℓ is convex

$\ell'(z) = \sigma(z) - 1$ and $\ell''(z) = \sigma(z)\sigma(-z) > 0$.

3.1 Surrogate risk bounds

For $(X, Y) \sim P$ and classifier $f = \operatorname{sign}(g)$, counting $g = 0$ as an error, the event $\{f(X) \neq Y\}$ implies $Yg(X) \leq 0$. For a nonnegative, nonincreasing loss ϕ this gives $\phi(Yg) \geq \phi(0)$ on errors and therefore

$$P(f(X) \neq Y) \leq \frac{\mathbb{E}[\phi(Yg(X))]}{\phi(0)}.$$

Logistic: $\phi(0) = \log 2$, so $R_{\log} < \varepsilon \Rightarrow P(f(X) = Y) > 1 - \varepsilon/\log 2$. Hinge $\max(0, 1 - u)$ and exponential e^{-u} have $\phi(0) = 1$. Minimizing a convex surrogate thus controls the 0–1 error, which itself cannot be optimized by gradients.

4 Multiclass Classification and Softmax

Motivation. With K classes the sigmoid generalizes to the softmax, which turns K scores into a probability vector. Its cross-entropy loss has the same clean gradient (prediction minus target) as in the binary case.

For Q classes, $W \in \mathbb{R}^{Q\times d}$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$ and

$$p_k(\mathbf{x}) = \operatorname{softmax}(\mathbf{s})_k = \frac{e^{s_k}}{\sum_r e^{s_r}}, \qquad J(W) = -\sum_{i,k} y_{ik} \log p_k(\mathbf{x}_i).$$

Scores $(2, 1, 0)$ give approximately $(0.665, 0.245, 0.090)$.

Theorem 4.7: softmax calculus

With $\operatorname{LSE}(\mathbf{s}) = \log \sum_r e^{s_r}$ and $\mathbf{p} = \operatorname{softmax}(\mathbf{s})$:

(i) $\operatorname{softmax}(\mathbf{s} + c\mathbf{1}) = \operatorname{softmax}(\mathbf{s})$, $\operatorname{LSE}(\mathbf{s} + c\mathbf{1}) = \operatorname{LSE}(\mathbf{s}) + c$; stable evaluation $\operatorname{LSE}(\mathbf{s}) = M + \log \sum e^{s_r - M}$, $M = \max s_r$.

(ii) $\nabla \operatorname{LSE} = \mathbf{p}$; Jacobian $\partial p_k / \partial s_m = p_k(\delta_{km} - p_m)$, i.e. $\nabla^2 \operatorname{LSE} = \operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top \succeq 0$.

(iii) $\nabla_{\mathbf{s}}(-\log p_y) = \mathbf{p} - \mathbf{e}_y$; for the whole data set $\nabla_W J = (P - Y)^\top X$ (rows: classes).

(iv) $\log \operatorname{softmax}(\mathbf{s}) = \mathbf{s} - \operatorname{LSE}(\mathbf{s})\mathbf{1}$ and log-softmax is idempotent: $\log \operatorname{softmax}(\log \operatorname{softmax}(\mathbf{s})) = \log \operatorname{softmax}(\mathbf{s})$.

(v) $\max_k s_k \le \operatorname{LSE}(\mathbf{s}) \le \max_k s_k + \log Q$.

Proof. (i) e^c cancels.

(ii) Quotient rule; $\mathbf{v}^\top(\operatorname{diag}\mathbf{p} - \mathbf{p}\mathbf{p}^\top)\mathbf{v} = \sum p_k v_k^2 - (\sum p_k v_k)^2$ is the variance of v_K with $K \sim \mathbf{p}$.

(iii) $-\log p_y = -s_y + \operatorname{LSE}(\mathbf{s})$.

(iv) With $\boldsymbol{\ell} = \log \operatorname{softmax}(\mathbf{s})$, $e^{\ell_j} = p_j$ so $\operatorname{LSE}(\boldsymbol{\ell}) = \log \sum p_j = 0$.

(v) $e^M \le \sum e^{s_r} \le Q e^M$. ∎

Adding the same vector to all rows of W leaves the softmax unchanged, so W is identifiable only up to such shifts.

5 Support Vector Machines

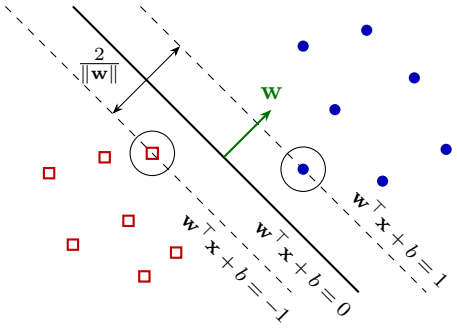
5.1 Geometry and the primal problem

Labels are $y_i \in \{\pm 1\}$, score $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The distance from $\mathbf{x}_0$ to $\{\mathbf{w}^\top \mathbf{x} + b = 0\}$ is $|\mathbf{w}^\top \mathbf{x}_0 + b|/\|\mathbf{w}\|$ (move along the unit normal: $\mathbf{x}_0 - t\mathbf{w}/\|\mathbf{w}\|$ lies on the plane iff $t = (\mathbf{w}^\top \mathbf{x}_0 + b)/\|\mathbf{w}\|$). Rescaling $(\mathbf{w}, b)$ by $c > 0$ changes nothing, so for separable data normalize the closest functional margin to 1:

Definition 4.8: hard-margin SVM

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \ge 1, \ i = 1, \dots, n.$$

The support planes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ lie at distance $1/\|\mathbf{w}\|$ (the margin) from the boundary; the slab width is $2/\|\mathbf{w}\|$.



Linear SVM: the margin between the hyperplanes $\mathbf{w}^\top\mathbf{x}+b=\pm1$ is $2/\|\mathbf{w}\|$; the circled points are support vectors.

5.2 Lagrangian, max–min inequality, weak duality

Theorem 4.9: max–min inequality

For any $F:\mathcal{X}\times\mathcal{Y}\to\mathbb{R}$, $\sup_x\inf_y F(x,y)\le\inf_y\sup_x F(x,y)$. Equality can fail: for $F=\begin{pmatrix}0&1\\1&0\end{pmatrix}$ (rows x, columns y) the left side is 0 and the right side 1.

Proof. For all x,y: $\inf_{y'}F(x,y')\le F(x,y)\le\sup_{x'}F(x',y)$. The left end does not depend on y and the right end not on x; take $\sup_x$ on the left, then $\inf_y$ on the right. ■

With $c_i=1-y_i(\mathbf{w}^\top\mathbf{x}_i+b)$ and $\alpha_i\ge0$, $L(\mathbf{w},b,\boldsymbol{\alpha})=\frac12\|\mathbf{w}\|^2+\sum_i\alpha_ic_i$. Since $\sup_{\alpha\ge0}\alpha c=0$ if $c\le0$ and $+\infty$ if $c>0$, $\sup_{\boldsymbol{\alpha}\ge\mathbf{0}}L$ equals the objective on feasible points and $+\infty$ elsewhere; hence $p^\star=\inf_{\mathbf{w},b}\sup_{\boldsymbol{\alpha}}L$. The dual is $d^\star=\sup_{\boldsymbol{\alpha}}\inf_{\mathbf{w},b}L$ and the max–min inequality gives **weak duality** $d^\star\le p^\star$: every dual value is a lower bound on the primal optimum. For the SVM (convex, strictly feasible) strong duality $d^\star=p^\star$ holds.

5.3 The dual problem

Theorem 4.10: SVM dual

$\nabla_{\mathbf{w}}L=\mathbf{0}\Rightarrow\mathbf{w}=\sum_i\alpha_iy_i\mathbf{x}_i$; finiteness of $\inf_b L$ requires $\sum_i\alpha_iy_i=0$. The dual is

$$\max_{\boldsymbol{\alpha}\ge\mathbf{0}}\ \sum_i\alpha_i-\frac12\sum_{i,j}\alpha_i\alpha_jy_iy_j\mathbf{x}_i^\top\mathbf{x}_j\quad\text{s.t.}\quad\sum_i\alpha_iy_i=0.$$

Complementary slackness $\alpha_i[1-y_i(\mathbf{w}^\top\mathbf{x}_i+b)]=0$: $\alpha_i>0$ only for support vectors, and $b=y_i-\mathbf{w}^\top\mathbf{x}_i$ for any such i.

Proof. b enters L linearly with coefficient $-\sum\alpha_iy_i$; if nonzero, $\inf_b=-\infty$. Substituting $\mathbf{w}$: $\frac12\|\mathbf{w}\|^2-\|\mathbf{w}\|^2+\sum\alpha_i=\sum\alpha_i-\frac12\|\mathbf{w}\|^2$. ■

Example 4.11: two points

$\mathbf{x}_1=(0,0)$, $y_1=-1$; $\mathbf{x}_2=(2,2)$, $y_2=+1$. The constraint forces $\alpha_1=\alpha_2=a$ and $\|\mathbf{x}_2\|^2=8$, so the dual is $2a-4a^2$, maximized at $a=1/4$. Then $\mathbf{w}=\frac14(2,2)=(1/2,1/2)$, $b=-1$ (from $\mathbf{x}_1$), the boundary is $x^{(1)}+x^{(2)}=2$, $\|\mathbf{w}\|=1/\sqrt2$, margin $\sqrt2$ (slab width $2\sqrt2$). Primal value $\frac12\|\mathbf{w}\|^2=1/4$ equals the dual value $2a-4a^2=1/4$.

Remark 4.12: soft margin and kernels

For non-separable data add slacks: $\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum \xi_i$ s.t. $y_i f(\mathbf{x}_i) \geq 1 - \xi_i$, $\xi_i \geq 0$, equivalently hinge loss $C\sum \max(0, 1 - y_i f(\mathbf{x}_i))$ plus ℓ_2. Its dual is identical except for the box constraint $0 \leq \alpha_i \leq C$. Replacing $\mathbf{x}_i^\top \mathbf{x}_j$ by a kernel gives the kernel SVM with score $\sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$. Logistic regression fits a likelihood; the SVM maximizes a geometric margin.

6 Selected exercises

Motivation. The exercises show how two-class softmax collapses to the logistic model, identify what the logistic loss estimates in the population, and read off the SVM margin from the dual variables.

Exercise 4.1 (Two-class softmax is logistic regression)

Let $Q = 2$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$, labels $y_i = y_{i1} \in \{0, 1\}$, $y_{i2} = 1 - y_i$, and $J(\mathbf{w}_1, \mathbf{w}_2)$ the softmax cross-entropy.
(a) Show that $p_1(\mathbf{x}) = \sigma((\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{x})$ and $J(\mathbf{w}_1, \mathbf{w}_2) = J_{\log}(\mathbf{w}_1 - \mathbf{w}_2)$, where $J_{\log}$ is the logistic loss of Section 2.
(b) Show that $\nabla_{\mathbf{w}_1} J = -\nabla_{\mathbf{w}_2} J$ and that the Hessian of J in $(\mathbf{w}_1, \mathbf{w}_2)$ is singular.
(c) For $\lambda > 0$ let $(\hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2)$ minimize $J + \frac{\lambda}{2}(\|\mathbf{w}_1\|^2 + \|\mathbf{w}_2\|^2)$. Show that $\hat{\mathbf{w}}_1 = -\hat{\mathbf{w}}_2$ and that $\hat{\mathbf{w}} = \hat{\mathbf{w}}_1 - \hat{\mathbf{w}}_2$ minimizes $J_{\log}(\mathbf{w}) + \frac{\lambda}{4}\|\mathbf{w}\|^2$.

Solution. [Supplement]

(a) Dividing numerator and denominator by e^{s_1},

$$p_1 = \frac{e^{s_1}}{e^{s_1} + e^{s_2}} = \frac{1}{1 + e^{-(s_1 - s_2)}} = \sigma((\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{x}), \qquad p_2 = 1 - p_1.$$

Hence $J = -\sum_i [y_i \log p_1(\mathbf{x}_i) + (1 - y_i) \log(1 - p_1(\mathbf{x}_i))]$, the logistic loss at $\mathbf{w} = \mathbf{w}_1 - \mathbf{w}_2$.

(b) By (a) and the chain rule, $\nabla_{\mathbf{w}_1} J = \nabla J_{\log}(\mathbf{w}_1 - \mathbf{w}_2)$ and $\nabla_{\mathbf{w}_2} J = -\nabla J_{\log}(\mathbf{w}_1 - \mathbf{w}_2)$. For any $\mathbf{v} \neq \mathbf{0}$, $t \mapsto J(\mathbf{w}_1 + t\mathbf{v}, \mathbf{w}_2 + t\mathbf{v})$ is constant, so with $\mathbf{u} = (\mathbf{v}, \mathbf{v})$ the Hessian $\mathcal{H}$ satisfies $\mathbf{u}^\top \mathcal{H} \mathbf{u} = 0$. J is convex (a convex function of a linear map), so $\mathcal{H} \succeq 0$, and then $\|\mathcal{H}^{1/2}\mathbf{u}\|^2 = 0$ gives $\mathcal{H}\mathbf{u} = \mathbf{0}$ with $\mathbf{u} \neq \mathbf{0}$.

(c) The penalized objective is strictly convex (convex plus strongly convex) and coercive, so it has a unique minimizer, characterized by

$$\nabla_{\mathbf{w}_1} J + \lambda \hat{\mathbf{w}}_1 = \mathbf{0}, \qquad \nabla_{\mathbf{w}_2} J + \lambda \hat{\mathbf{w}}_2 = \mathbf{0}.$$

Adding and using (b) gives $\lambda(\hat{\mathbf{w}}_1 + \hat{\mathbf{w}}_2) = \mathbf{0}$, so $\hat{\mathbf{w}}_1 = -\hat{\mathbf{w}}_2 = \hat{\mathbf{w}}/2$. Restricted to pairs $(\mathbf{w}/2, -\mathbf{w}/2)$ the objective equals $J_{\log}(\mathbf{w}) + \frac{\lambda}{2} \cdot 2\|\mathbf{w}/2\|^2 = J_{\log}(\mathbf{w}) + \frac{\lambda}{4}\|\mathbf{w}\|^2$. The minimizer over all pairs lies on this set, so $\hat{\mathbf{w}}$ minimizes the restricted objective. The penalty thus picks the symmetric representative among all equivalent softmax parameters. ■

Exercise 4.2 (What logistic regression estimates)

Let $(X, Y) \sim P$ with $Y \in \{\pm 1\}$ and $\eta(\mathbf{x}) = P(Y = +1 \mid X = \mathbf{x}) \in (0, 1)$. For measurable g let $R(g) = \mathbb{E}[\ell(Y g(X))]$ with $\ell(z) = \log(1 + e^{-z})$.
(a) For $\eta \in (0, 1)$ show that $C_\eta(t) = \eta\, \ell(t) + (1 - \eta)\ell(-t)$ is strictly convex with unique minimizer $t^\star = \operatorname{logit}(\eta)$.

(b) Deduce that $g^\star(\mathbf{x}) = \operatorname{logit}\eta(\mathbf{x})$ minimizes R over all measurable g, and that $\operatorname{sign} g^\star$ is the Bayes classifier (predict $+1$ iff $\eta > \frac{1}{2}$).
(c) Show that $C_\eta(t^\star) = -\eta\log\eta - (1-\eta)\log(1-\eta)$, so $R(g^\star)$ is the expected binary entropy of $\eta(X)$.

Solution. [Supplement]

(a) By the theorem "ℓ is convex", $\ell'(z) = \sigma(z) - 1 = -\sigma(-z)$ and $\ell'' > 0$. Hence $\frac{d}{dt}\ell(-t) = \sigma(t)$ and

$$C'_\eta(t) = -\eta\,\sigma(-t) + (1-\eta)\sigma(t) = -\eta(1-\sigma(t)) + (1-\eta)\sigma(t) = \sigma(t) - \eta.$$

$C''_\eta(t) = \sigma'(t) > 0$, so C_η is strictly convex and its unique stationary point, $\sigma(t) = \eta$, i.e. $t = \operatorname{logit}(\eta)$, is the unique global minimizer.

(b) Conditioning on X, $\mathbb{E}[\ell(Yg(X)) \mid X] = C_{\eta(X)}(g(X))$, so $R(g) = \mathbb{E}\,C_{\eta(X)}(g(X))$. By (a), $C_{\eta(X)}(g(X)) \geq C_{\eta(X)}(g^\star(X))$ pointwise; taking expectations gives $R(g) \geq R(g^\star)$, where $R(g^\star) \leq C_\eta(0) = \log 2$ is finite and $g^\star$ is measurable as a composition of measurable maps. Finally $\operatorname{logit}\eta > 0$ iff $\eta > \frac{1}{2}$.

(c) From $\sigma(t^\star) = \eta$: $\ell(t^\star) = -\log\sigma(t^\star) = -\log\eta$ and $\ell(-t^\star) = -\log\sigma(-t^\star) = -\log(1-\eta)$. Substituting into C_η gives the entropy. So minimizing the logistic loss estimates the conditional log-odds, not just the decision; this is why its outputs can be read as probabilities. ■

Exercise 4.3 (The margin from the dual variables)

Let the data be linearly separable with both labels present, let $(\mathbf{w}^\star, b^\star)$ solve the hard-margin primal and $\boldsymbol{\alpha}^\star$ the dual, with the KKT conditions of the dual theorem in force.
(a) Show that $\mathbf{w}^\star$ is unique, $\mathbf{w}^\star \neq \mathbf{0}$, and $b^\star$ is uniquely determined.
(b) Show that $\|\mathbf{w}^\star\|^2 = \sum_i \alpha_i^\star$; hence the margin is $\left(\sum_i \alpha_i^\star\right)^{-1/2}$ and the primal and dual values coincide.
(c) Check (b) on the two-point example.

Solution. [Supplement]

(a) *Step 1 (uniqueness of $\mathbf{w}^\star$).* The feasible set F is an intersection of half-spaces in $(\mathbf{w}, b)$, hence convex. Suppose $(\mathbf{w}_1, b_1), (\mathbf{w}_2, b_2) \in F$ are both optimal with value $p^\star$ and $\mathbf{w}_1 \neq \mathbf{w}_2$. Their midpoint is feasible and, by the parallelogram identity,

$$\tfrac{1}{2}\left\|\tfrac{\mathbf{w}_1+\mathbf{w}_2}{2}\right\|^2 = \tfrac{1}{4}\|\mathbf{w}_1\|^2 + \tfrac{1}{4}\|\mathbf{w}_2\|^2 - \tfrac{1}{8}\|\mathbf{w}_1 - \mathbf{w}_2\|^2 < p^\star,$$

a contradiction.

Step 2 ($\mathbf{w}^\star \neq \mathbf{0}$). With $\mathbf{w} = \mathbf{0}$ the constraints read $y_i b \geq 1$; a point with $y_i = +1$ forces $b \geq 1$ and one with $y_i = -1$ forces $b \leq -1$. So $\mathbf{w}^\star \neq \mathbf{0}$.

Step 3 ($b^\star$). Since $\mathbf{w}^\star = \sum_i \alpha_i^\star y_i \mathbf{x}_i \neq \mathbf{0}$, some $\alpha_i^\star > 0$. Complementary slackness gives $y_i(\mathbf{w}^{\star\top}\mathbf{x}_i + b^\star) = 1$, and $y_i^2 = 1$ gives $b^\star = y_i - \mathbf{w}^{\star\top}\mathbf{x}_i$, a function of $\mathbf{w}^\star$ alone.

(b) Summing complementary slackness over i and using $\sum_i \alpha_i^\star y_i = 0$,

$$\sum_i \alpha_i^\star = \sum_i \alpha_i^\star y_i(\mathbf{w}^{\star\top}\mathbf{x}_i + b^\star) = \mathbf{w}^{\star\top}\sum_i \alpha_i^\star y_i \mathbf{x}_i + b^\star \sum_i \alpha_i^\star y_i = \|\mathbf{w}^\star\|^2.$$

The margin is $1/\|\mathbf{w}^\star\|$. The primal value is $\frac{1}{2}\|\mathbf{w}^\star\|^2$, and the dual value is $\sum_i \alpha_i^\star - \frac{1}{2}\|\mathbf{w}^\star\|^2 = \frac{1}{2}\|\mathbf{w}^\star\|^2$.

(c) There $\alpha_1 = \alpha_2 = \frac{1}{4}$, so $\sum_i \alpha_i = \frac{1}{2} = \|(1/2, 1/2)\|^2$ and the margin is $(1/2)^{-1/2} = \sqrt{2}$, as found directly. ■